

Notes: Kiyotaki and Moore (JPE, 1997)

Credit Cycles

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1 Big picture

- **Question.** How do credit constraints tied to collateral values interact with asset prices to amplify, propagate, and spread otherwise small shocks through the macroeconomy?
- **Setting.** The paper builds a dynamic general-equilibrium model in which durable assets — especially land — play a **dual role**: they are both factors of production and collateral for loans. Some producers are financially constrained and highly leveraged; others are unconstrained.
- **Core challenge.** A temporary shock to productivity or net worth should ordinarily have only temporary effects. The paper asks why, in a credit economy, such a shock can instead produce **large and persistent** movements in asset prices, borrowing, investment, and output.
- **Main theoretical contribution.** The paper identifies a **dynamic multiplier** that works through the interaction between borrowers' net worth and collateral prices:
 - a negative shock lowers constrained firms' net worth,
 - they cut land demand,
 - the land price falls,
 - the fall in land price reduces collateral value and net worth even more,
 - this worsens future borrowing capacity and keeps depressing investment.

Persistence and amplification therefore reinforce one another.

- **Main results.**
 - In the **basic model**, the dynamic multiplier on land prices is much stronger than the static one; the paper shows that the extra movement in land prices caused by the dynamic multiplier is $1/(R - 1)$ **times** the static movement, where $R - 1$ is the net real interest rate.
 - Because constrained farmers have a higher marginal product of land than unconstrained gatherers, reallocation of land away from constrained agents causes a **first-order drop in aggregate output**.
 - In the **full model**, adding an uncollateralizable reproducible asset (trees) and lumpy investment opportunities produces **more persistence, hump-shaped responses**, and even **damped cycles**.
 - In a **spillover extension**, a shock to one sector moves the common land price and therefore transmits to other sectors, generating comovement.
- **Why it matters.** The paper provides one of the classic foundations for the modern macro-finance view that balance sheets and collateral values are not passive outcomes. They actively shape macroeconomic dynamics.
- **Simulation takeaway.** In the calibrated quarterly full model, a 1% temporary productivity shock raises the land price by about **0.37%** on impact. Farmers' landholdings and debt rise by about **0.10%** and **0.13%** on impact, then build further and peak roughly **seven quarters later**. The cycle lasts about **10 years**.
- **Economic takeaway.** The paper's key idea is that when credit limits depend on collateral values, asset prices become a **propagation mechanism**, not just a reflection of fundamentals.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the key objects are

- There is **no paper-specific empirical dataset**. The important objects are the primitives of the model: agents, technologies, collateral constraints, asset prices, borrowing, and the dynamic laws of motion.
- The paper proceeds in three layers:
 - a **basic model** with land and fruit,
 - a **full model** that adds trees and lumpy investment,
 - a **spillover extension** with two sectors linked by a common land market.

2.2 Agents, goods, and markets in the basic model

- **Goods.**
 - a **durable asset**: land, fixed in aggregate supply at K ,
 - a **nondurable commodity**: fruit, which cannot be stored.
- **Agents.**
 - **Farmers**, population normalized to 1: potentially productive but credit constrained.
 - **Gatherers**, population m : unconstrained agents with decreasing-returns production.
- **Markets.**
 - a competitive spot market for land, with price q_t ,
 - a one-period credit market, with gross interest rate R_t .
- **Preferences.** Both types are risk neutral, but farmers are more impatient than gatherers: $\beta < \beta'$.

2.3 Farmers' technology and the source of credit constraints

- A farmer using k_t units of land at date t produces at $t + 1$

$$y_{t+1} = (a + c)k_t.$$

- Of this output, only ak_t is tradable. The rest, ck_t , is “bruised” fruit that can be consumed by the farmer but not traded.
- The crucial institutional assumption is that the farmer’s human capital is **inalienable**. If the farmer walks away, creditors seize only the land, not the inside value generated by his labor.
- Therefore land can serve as collateral, but borrowing is limited by the market value of land:

$$Rb_t \leq q_{t+1}k_t.$$

- **Interpretation.** The paper generates credit limits from limited commitment, not from an ad hoc borrowing cap.

2.4 Gatherers and the residual supply of land

- Gatherers have a decreasing-returns technology $G(k'_t)$ and face no credit constraint.
- Since they are unconstrained, their demand for land is pinned down by the **user cost** of land rather than by net worth.
- This gives the residual supply of land to farmers. The elasticity of that residual supply, denoted η , is central for the size of amplification.

2.5 Key endogenous objects

- q_t : land price.
- $u_t \equiv q_t - q_{t+1}/R$: user cost of land and, equivalently, the down payment per unit of land.
- K_t : aggregate landholding of farmers.
- B_t : aggregate borrowing of farmers.
- Net worth of constrained producers: current tradable output plus land value minus debt repayment.

2.6 What changes in the full model

- The full model introduces **trees**, a reproducible asset planted on land.
- Trees depreciate at rate $1 - \lambda$ and require fruit investment at unit cost ϕ .
- Trees are **specific** to the farmer who planted them and therefore cannot be collateralized.
- Only a fraction π of farmers receive an **investment opportunity** in any period. This creates lumpy investment and separates aggregate debt from aggregate landholdings.

2.7 What changes in the spillover model

- The paper adds two farming sectors, each producing a differentiated type of fruit.
- Both sectors share the same land market, so movements in the land price transmit shocks across sectors even when the original disturbance is sector-specific.

3 Models and empirical specifications (all equations + interpretation)

3.1 Basic model: preferences and production

Farmers and gatherers have linear expected utility:

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s x_{t+s} \quad \text{and} \quad \mathbb{E}_t \sum_{s=0}^{\infty} (\beta')^s x'_{t+s}. \quad (1)$$

The farmer's production function is

$$y_{t+1} = F(k_t) \equiv (a + c)k_t. \quad (2)$$

Only ak_t is tradable; ck_t is nontradable and consumed by the farmer.

Interpretation.

- Farmers have constant returns to scale in land, so absent borrowing constraints they would want to expand until their marginal productivity is driven down.
- The split between tradable and nontradable fruit ensures that farmers consume something but still want to lever up aggressively.

3.2 Borrowing constraint and farmer flow of funds

The farmer's debt must be secured by land:

$$Rb_t \leq q_{t+1}k_t. \quad (3)$$

The flow-of-funds constraint is

$$q_t(k_t - k_{t-1}) + Rb_{t-1} + x_t - ck_{t-1} = ak_{t-1} + b_t. \quad (4)$$

Interpretation.

- The farmer rolls over debt, repays old debt, consumes, and buys new land using current tradable output plus new borrowing.
- Because future land value enters the collateral constraint, the expected future price q_{t+1} affects current borrowing capacity.

3.3 Gatherers, user cost, and land market clearing

Gatherers produce according to

$$y'_{t+1} = G(k'_t), \quad G' > 0, G'' < 0. \quad (5)$$

Since gatherers are unconstrained, they choose land such that

$$\frac{1}{R}G'(k'_t) = u_t, \quad u_t \equiv q_t - \frac{q_{t+1}}{R}. \quad (11)$$

Using market clearing $K_t + mk'_t = K$, the land market equilibrium condition is

$$u_t = q_t - \frac{q_{t+1}}{R} = u(K_t), \quad u(K) \equiv \frac{1}{R}G'\left(\frac{1}{m}(K - K_t)\right). \quad (12)$$

Interpretation.

- The user cost u_t is both the gatherers' opportunity cost of holding land and the farmers' required down payment per unit of land.
- A higher farmer land demand K_t must be matched by a higher user cost to crowd gatherers out of the land market.

3.4 Farmer optimality, aggregation, and the basic steady state

Near the steady state, farmers borrow up to the limit and consume only the nontradable fruit. Their optimal landholding is

$$k_t = \frac{(a + q_t)k_{t-1} - Rb_{t-1}}{q_t - q_{t+1}/R}. \quad (7)$$

Aggregating across farmers gives

$$K_t = \frac{(a + q_t)K_{t-1} - RB_{t-1}}{u_t}, \quad (9)$$

$$B_t = \frac{q_{t+1}}{R}K_t. \quad (10)$$

In steady state,

$$\frac{R-1}{R}q^* = u^* = a, \quad (13a)$$

$$\frac{1}{R}G'\left(\frac{K - K^*}{m}\right) = u^*, \quad (13b)$$

$$B^* = \frac{a}{R-1}K^*. \quad (13c)$$

Interpretation.

- Equation (7) says farmers use net worth to finance the down payment on land.
- The steady state features **high leverage**: debt is proportional to the value of land.
- Compared with the first best, farmers hold too little land because they are borrowing constrained. Since farmers have the higher marginal product of land, this creates a first-order output loss.

3.5 Shock propagation in the basic model: the dynamic multiplier

Suppose the economy starts in steady state and is hit by a small temporary productivity shock Δ at date t . Linearizing around the steady state, the farmer landholding dynamics satisfy

$$\left(1 + \frac{1}{\eta}\right) \hat{K}_t = \Delta + \frac{R}{R-1} \hat{q}_t, \quad (15a)$$

$$\left(1 + \frac{1}{\eta}\right) \hat{K}_{t+s} = \hat{K}_{t+s-1}, \quad s \geq 1. \quad (15b)$$

Land price equals the discounted sum of future user costs, which implies

$$\hat{q}_t = \frac{R-1}{\eta R} \sum_{s=0}^{\infty} R^{-s} \hat{K}_{t+s} = \frac{R-1}{\eta R} \frac{1}{1 - \eta/[R(1+\eta)]} \hat{K}_t. \quad (16)$$

Solving gives

$$\hat{q}_t = \frac{1}{\eta} \Delta, \quad (17)$$

$$\hat{K}_t = \left(1 + \frac{1}{(1+\eta)(R-1)}\right) \Delta. \quad (18)$$

If there were *no* dynamic multiplier and future land prices were held fixed at q^* , the response would instead be

$$\hat{q}_t^{\text{static}} = \frac{R-1}{R\eta} \Delta, \quad \hat{K}_t^{\text{static}} = \Delta. \quad (17'-18')$$

Interpretation.

- The direct effect of the shock raises current net worth.
- The indirect effect is much more powerful: a higher land price boosts collateral value, which further raises borrowing and land demand.
- The dynamic multiplier is stronger than the static one because current land prices capitalize **future** declines in user costs.
- This is the core financial-accelerator mechanism of the paper.

3.6 Full model: trees as uncollateralizable capital

The full model adds trees. A fraction $1 - \lambda$ of trees dies each period, and investing farmers must spend $\phi(k_t - \lambda k_{t-1})$ units of fruit to replant and expand. The farmer's flow-of-funds constraint becomes

$$q_t(k_t - k_{t-1}) + \phi(k_t - \lambda k_{t-1}) + Rb_{t-1} + x_t - ck_{t-1} = ak_{t-1} + b_t. \quad (20)$$

If the farmer has an investment opportunity, optimal land choice is

$$k_t = \frac{(a + q_t + \lambda\phi)k_{t-1} - Rb_{t-1}}{\phi + q_t - q_{t+1}/R}. \quad (21)$$

If the farmer does *not* have an investment opportunity, then

$$k_t \leq \lambda k_{t-1}, \quad \text{and in equilibrium } k_t = \lambda k_{t-1}. \quad (22)$$

Aggregating across investing and noninvesting farmers yields

$$K_t = (1 - \pi)\lambda K_{t-1} + \pi \frac{(a + q_t + \lambda\phi)K_{t-1} - RB_{t-1}}{\phi + q_t - q_{t+1}/R}, \quad (23)$$

$$B_t = RB_{t-1} + q_t(K_t - K_{t-1}) + \phi(K_t - \lambda K_{t-1}) - aK_{t-1}. \quad (24)$$

Interpretation.

- Trees are productive but cannot be pledged as collateral. This reduces leverage relative to the basic model.
- Because investment now requires both land and trees, the required down payment is larger, so the impact response is less explosive.
- But precisely because trees are uncollateralizable, shocks become more persistent: quantities move less on impact, and more of the adjustment shows up in asset prices.

3.7 Lumpy investment opportunities and cycles

Only a fraction π of farmers can invest in each period. This breaks the tight aggregate link between debt and landholding. On the stable manifold, the land price can be written as

$$\hat{q}_{t+s} = \mu_K \hat{K}_{t+s} - \mu_B \hat{B}_{t+s}, \quad (26)$$

with $\mu_K > 0$ and $\mu_B > 0$.

Interpretation.

- Higher landholdings raise the current land price because they predict higher future demand for collateral.
- Higher debt lowers the current land price because it implies future debt overhang.
- The interaction between landholdings (the “prey”) and debt (the “predator”) can generate damped oscillations.
- The paper’s simulations show that land price moves first, then landholdings and debt build up gradually, producing a hump-shaped cycle.

3.8 Spillover extension with two sectors

The paper then considers two farming sectors, indexed by $i = 1, 2$, producing differentiated fruit. Relative demand implies a sectoral fruit price

$$p_{it} = (a_i K_{i,t-1})^{-e} [(a_1 K_{1,t-1})^{1-e} + (a_2 K_{2,t-1})^{1-e}]^{e/(1-e)}. \quad (31)$$

Sector- i farmers choose land according to

$$K_{it} = \frac{(a_i p_{it} + q_t) K_{i,t-1} - R B_{i,t-1}}{q_t - q_{t+1}/R}, \quad (32)$$

with debt

$$B_{it} = \frac{q_{t+1}}{R} K_{it}, \quad (33)$$

and a common land market satisfying

$$q_t - \frac{q_{t+1}}{R} = u(K_{1t} + K_{2t}). \quad (34)$$

Interpretation.

- A positive shock to sector 1 raises the common land price.
- Since sector-2 farmers also hold land, they get a capital gain and higher collateral value even though their own productivity did not improve.
- Thus a sector-specific shock spills over through the shared collateral market.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the mechanism work

- The paper does **not** rely on arbitrary reduced-form borrowing limits. The constraint is derived from limited commitment: farmers can walk away with their human capital, so debt must be secured.
- The same durable asset — land — is both:
 - an input into production,
 - collateral determining borrowing capacity.

This dual role is what lets asset prices affect real activity.

4.2 Why amplification is stronger than in a static collateral story

- A purely static story says lower net worth cuts current asset demand and lowers the current asset price.
- Kiyotaki–Moore’s key addition is that today’s asset price depends on expected **future** user costs, and future user costs depend on future borrowing capacity.
- Therefore a current shock affects all future dates, and the capitalization of those future effects feeds back immediately into the current price.

4.3 Why persistence arises endogenously

- Farmers' future investment depends on future net worth.
- Future net worth depends on current production, current debt, and the collateral value of land.
- Hence even a temporary shock changes the state variables that matter tomorrow, so the economy does not jump back immediately.

4.4 What the full model adds

- The addition of trees weakens leverage but strengthens persistence.
- The introduction of lumpy investment opportunities means not all constrained agents respond at once, which naturally creates delayed peaks and cyclical dynamics.
- This is why the full model produces a much more realistic hump-shaped adjustment than the stark basic model.

4.5 Relation to the literature

- Relative to **Bernanke and Gertler (1989)**, this paper adds a much sharper role for **asset prices** as the link between net worth today and credit capacity tomorrow.
- Relative to **Shleifer and Vishny (1992)**, it turns a largely static fire-sale idea into a dynamic model where amplification and persistence reinforce one another.
- The paper anticipates a large macro-finance literature in which collateral values, net worth, and leverage shape aggregate fluctuations.

4.6 Caveats highlighted by the authors

- The main text studies unexpected temporary shocks in an otherwise deterministic environment.
- The paper does not endogenize who is constrained and when in the baseline.
- It abstracts from contingent debt contracts tied to aggregate states and from informational frictions on the lenders' side.
- The appendix shows that the main mechanism does not depend on the paper's convenient linear-preferences shortcut by presenting an overlapping-generations version with more orthodox preferences.

5 Tables and figures

5.1 Figure 1: The credit-cycle mechanism

- Figure 1 is the conceptual heart of the paper. It diagrams the sequence:
shock $\rightarrow$ net worth falls $\rightarrow$ asset demand falls $\rightarrow$ user cost falls $\rightarrow$ asset price falls.
- The figure separates a **static multiplier** from a **dynamic multiplier**.
 - Static: lower current demand lowers the current price.
 - Dynamic: lower future demand lowers future user costs, which lowers the current price even more through capitalization.
- Read the figure as a map of the paper's logic: the main novelty is not just leverage, but leverage combined with asset pricing.

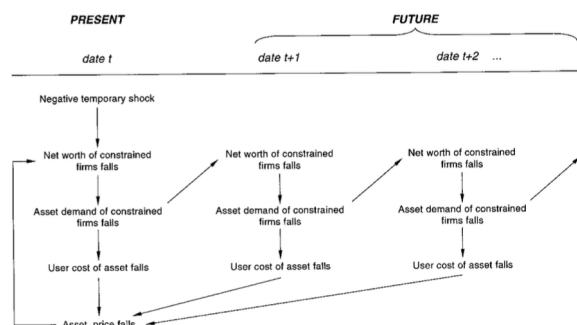


FIG. 1

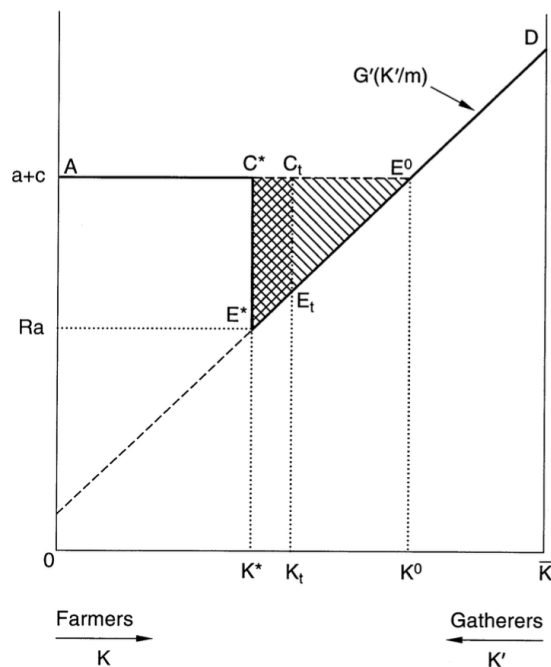


FIG. 2

5.2 Figure 2: Steady-state misallocation under credit constraints

- Figure 2 compares the constrained steady state with the first best.
- In the first best, the marginal product of land is equalized across farmers and gatherers.
- In the constrained equilibrium, farmers hold too little land, so their marginal product of land is **above** gatherers' marginal product.
- The triangle between the two allocations is the output loss due to credit constraints.

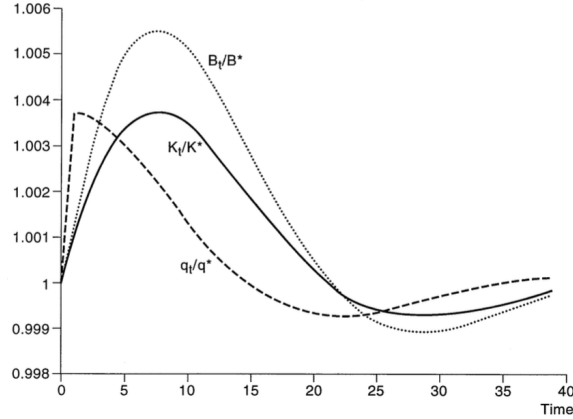


FIG. 3

- This figure is important because it shows that the model is not only about volatility. It is also about **miscallocation in levels**.

5.3 Figure 3: Simulation dynamics in the full model

- Figure 3 simulates the full model after a 1% temporary productivity shock.
- On impact:
 - land price rises by about **0.37%**,
 - farmers' landholding rises by about **0.10%**,
 - farmers' debt rises by about **0.13%**.
- The responses then build up further:
 - landholdings peak around **0.37%**,
 - debt peaks around **0.55%**,
 - both peaks occur roughly **seven quarters** after the shock.
- The overall cycle lasts around **40 quarters**, or **10 years**. Land price leads the cycle, while borrowing and landholding follow.
- This figure is the paper's numerical proof that the mechanism can generate realistic persistence and comovement, not just a one-period amplification.

6 Short summary

- **Method.** Build a dynamic equilibrium model in which land is both productive and collateralizable, so borrowing capacity depends on asset prices.
- **Mechanism.** A temporary shock changes constrained firms' net worth, which changes land demand, land prices, and hence future borrowing capacity. The asset price response capitalizes these future effects and amplifies the current shock.

- **Main result.** In the basic model, the dynamic collateral-price multiplier is much larger than the static one. In the full model, uncollateralizable capital and lumpy investment opportunities generate persistent hump-shaped responses and damped cycles.
- **Broader implication.** Credit constraints can turn temporary sectoral or productivity shocks into long-lived macroeconomic fluctuations and can transmit these shocks across sectors through a common collateral market.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that collateral constraints and asset prices interact in a way that can generate **amplification**, **persistence**, and **spillovers**.
- It is not necessary to assume large primitive shocks. Small temporary shocks can produce large movements in credit, asset prices, investment, and output because balance-sheet effects are endogenous.
- The paper also shows that this mechanism is not purely contemporaneous. It is inherently intertemporal.

7.2 Economic intuition

- When borrowers are leveraged, a fall in the price of collateral destroys net worth disproportionately.
- Since future borrowing depends on future net worth, this reduces future demand for the collateral asset.
- But the current asset price is the discounted value of future user costs. So the expectation of weak future demand lowers the current price today.
- That is the circular logic of the credit cycle: weaker balance sheets lower collateral values, and lower collateral values weaken balance sheets.

7.3 Takeaway

This paper is a classic because it shows in the cleanest possible way how **credit markets and asset markets co-determine each other**. The central lesson is that collateral prices are not merely signals of future conditions; they are part of the causal mechanism that creates those conditions. Once durable assets back debt, balance-sheet shocks become macro shocks.